CHEMISTRY 2

ACADEMIC

Grade 11/12

Curriculum Guide

CHEMISTRY 2

Prerequisite: Chemistry 1

Course Description:

This course builds on the foundational concepts in Chemistry 1 by enabling learners to examine chemical systems using quantitative relationships and experimental investigations. Learners explore the behavior of gases through the postulates of Kinetic Molecular Theory, relationships described by the Gas Laws, and the Ideal Gas Equation to explain how gas particles move and respond to changes in pressure, volume, temperature, and amount of gas. Learners perform calculations on Gas Laws using real-world problems. The learners also explain properties of solutions, factors affecting solubility, calculating solution concentrations, and discuss colligative properties.

Moreover, learners gain a fundamental understanding of the Law of Conservation of Mass as the basis of mass relationships in chemical reactions, write balanced chemical equations, and perform calculations to determine quantities of reactants and products. They also determine the limiting reagent and theoretical yield in a given chemical reaction. Additionally, they examine factors influencing reaction rates and analyze experimental data to determine reaction order and the Rate Law. Furthermore, they learn energy changes in chemical reactions through thermochemistry by examining calorimetry, enthalpy changes, and bond energies, describing entropy, and integrating these concepts with Gibbs free energy to explain the spontaneity of chemical processes.

Collectively, all the lessons emphasize applying chemistry concepts through a systems-thinking approach to address environmental, community, and industrial challenges. Learners explore sustainable practices related to Green Chemistry and connect their lesson understanding to the UN Sustainable Development Goals.

Elective: Academic

Time Allotment: 80 hours for one term

CONTENT	CONTENT STANDARDS *The learners demonstrate understanding that…*	LEARNING COMPETENCIES *The learners…*
Gases and Solutions		
1. Gases	1. the Kinetic Molecular Theory as a model for explaining gas behavior; 2. the quantitative relationships among pressure, volume, temperature, and amount of gas as described by the Gas Laws and the ideal gas equation;	1. explain the behavior of gases using the Kinetic Molecular Theory (KMT);
		2. conduct investigations on real gas experiments to show Boyle's Law, Charles' Law, Gay-Lussac's law, and Avogadro's Law;
		3. use Gas Law equations including ideal gas equation, to determine pressure, volume, temperature, or number of moles of a gas;
		4. determine the mole fraction and partial pressure of gases in a mixture using Dalton's Law of Partial Pressure;
		5. analyze applications of Gas Laws in phenomena m observed in daily activities, industrial practices, or applications related to addressing sustainability challenges;
2. Liquids and Solutions	3. properties of liquids and solutions are determined by intermolecular forces and solute–solvent interactions and can be described quantitatively;	6. explain the properties of solutions, such as solubility and viscosity in terms of intermolecular forces;
		7. describe how the temperature, pressure (for gases), and the nature of solute and solvent influence solubility, and relate these factors to intermolecular interactions;
		8. calculate solution concentrations, such as percent composition, molality, and molarity;
		9. explain effect of concentration on the colligative properties of solutions and apply these concepts to real-life contexts such as food, environmental, or industrial systems; and
		10. conduct experiments of colligative properties such as boiling point elevation and freezing point depression.

<table>
<tr><th>CONTENT</th><th>CONTENT STANDARDS

The learners demonstrate understanding that…</th><th>LEARNING COMPETENCIES

The learners…</th></tr>
<tr><td colspan="3">Performance Standards

The learners can explain gas behavior using the Kinetic Molecular Theory and conduct investigations on Gas Laws. They discuss properties of solutions and the factors affecting solubility as well as conduct experiments on colligative properties. The learners analyze how Gas Laws and solubility can be related to real-world scenarios and can be used to support sustainable development.</td></tr>
<tr><td colspan="3">Suggested Performance Tasks
Conduct an investigation to determine the relationship between gas variables (e.g., pressure and volume or temperature and volume), present data using graphs, and explain the results using gas laws.
Conduct experiments on colligative properties to explain how these concepts are applied in a real-world process (e.g., preparation of solutions or chemical production).</td></tr>
<tr><td colspan="3" align="center">Chemical Reactions, Stoichiometry, and Thermodynamics</td></tr>
<tr><td rowspan="5">3. Chemical Reactions and Stoichiometry</td><td rowspan="5">4. chemical reactions follow the Law of Conservation of Mass and can be represented and analyzed quantitatively;

5. mass relationships in chemical reactions help to understand how to calculate the amounts of reactants and products involved in chemical reactions observed in daily life activities, environmental science, and industrial processes;</td><td>1. determine different types of chemical reactions in living organisms, environment, and industries;</td></tr>
<tr><td>2. write the balanced chemical equations of various chemical reactions in real-life scenarios by applying the Law of Conservation of Mass;</td></tr>
<tr><td>3. calculate mass, mole, and number of particles in a given chemical reaction such as in respiration, photosynthesis, and environmental and industrial reactions;</td></tr>
<tr><td>4. determine the limiting reagent in a chemical reaction and use it to calculate the theoretical yield of products;</td></tr>
<tr><td>5. analyze the environmental effects of chemical reactions by examining their efficiency and byproducts within interconnected systems, and propose sustainable strategies based on Green Chemistry to minimize negative impacts;</td></tr>
</table>

CONTENT	CONTENT STANDARDS *The learners demonstrate understanding that…*	LEARNING COMPETENCIES *The learners…*
4. Chemical Kinetics	6. reaction rates depend on molecular collisions and can be explained using collision theory and rate laws;	6. describe the factors affecting reaction rates using laboratory experiments, simulations, or any digital technologies particularly in relation to everyday life activities and socio-scientific issues;
		7. use experimental data to determine rate law expressions and reaction order;
5. Chemical Thermodynamics	7. energy changes in chemical reactions can be described as endothermic and exothermic processes; and 8. thermodynamic principles explain energy transfer, disorder, and spontaneity of chemical processes.	8. conduct practical investigations to determine temperature changes in endothermic and exothermic reactions;
		9. explain enthalpy changes in chemical reactions by using calorimetry and heat capacity calculations, and relate energy changes to bond breaking and formation in accordance with the Law of Conservation of Energy;
		10. explain Gibbs free energy and relate it to the spontaneity of chemical processes; and
		11. analyze how the principles of thermodynamics are applied in everyday activities and in designing solutions that promote the Sustainable Development Goals (SDGs) and Green Chemistry.

Performance Standards

The learners can describe different types of chemical reactions, balance a chemical equation, and explain the Law of Conservation of Mass. Learners perform stoichiometric calculations to determine quantities of reactants and products. They analyze reaction rates using Collision Theory, explain the factors affecting reaction rate, and analyze and experimental data to determine Rate Law. They explain energy changes in chemical reactions using thermodynamic principles and conduct investigations to describe endothermic and exothermic processes. Learners apply the lessons to explain industrial and environmental processes while relating them to Green Chemistry and UN Sustainable Development Goals.

Suggested Performance Tasks

- Create an infographic on a selected chemical reaction that is experienced in real-life settings or those emphasizing its application in environmental chemistry, pharmaceuticals, and industrial processes. They can also share innovation or

CONTENT	CONTENT STANDARDS *The learners demonstrate understanding that…*	LEARNING COMPETENCIES *The learners…*
proposed possible innovations aligned with green chemistry that could be used to achieve UN Sustainable Development Goals. • Design and conduct an investigation to determine how temperature, concentration, surface area, or catalysts affect reaction rates. • Conduct a calorimetry experiment to determine whether a reaction is endothermic or exothermic, analyze temperature changes, and explain results using energy concepts.		

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For the Chemistry curriculum to be effective, we need:

***Continuous Training of Teachers**
***Development of Laboratory Activities**
***Hiring of Laboratory Technician**